

Detecting Phishing URLs using URL-based and Webpage-based Features

Justen Mao,
Lovejoy High School, 2350 Estates Pkwy, Lucas, TX 75002

1. Abstract

The increasing number of devices connected to the internet, including smartphones, IoT devices, and cloud networks, has expanded the attack surface for cybercriminals. Among cyber threats, phishing attacks are particularly effective because they exploit human vulnerabilities rather than system weaknesses. In a phishing attack, users are deceived into providing sensitive information, such as login credentials or financial data, which can then be used for further sophisticated attacks. Detecting phishing URLs before a user accesses a website is critical to prevent such breaches. This study investigates machine learning approaches to classify URLs as phishing or legitimate using URL-based and webpage-based features. Using the PhiUSIIL dataset, which contains 235,795 labeled samples with 54 features, models including Random Forest, Decision Tree, K-Nearest Neighbors, Logistic Regression, and Multi-Layer Perceptron were evaluated. Results show that URL-based features alone provide extremely high predictive power, with Random Forest achieving 99.922 percent accuracy and 100 percent recall, meaning all phishing URLs were correctly flagged before a webpage is accessed. Webpage features also contributed useful information, though with slightly lower accuracy of 99.336 percent. These findings highlight the feasibility of efficient, real-time phishing detection systems that proactively prevent users from visiting malicious websites.

2. Introduction

Phishing attacks remain a critical threat to online security, tricking users into disclosing sensitive information such as passwords, credit card numbers, or personal data. The research question addressed in this study is how machine learning models can accurately classify URLs as phishing or legitimate using URL and webpage features. Effective phishing detection is essential because it can prevent users from visiting malicious websites and reduce risks of financial loss and identity theft.

This project focuses on a supervised binary classification problem, where URLs are labeled as phishing (0) or legitimate (1). The data are numerical and categorical features extracted from the PhiUSIIL dataset, including URL length, domain properties, obfuscation indicators, HTML structure, and interactive webpage elements. The output is a binary label predicting whether a URL is phishing. Multiple machine learning models, including Logistic Regression, Decision Tree, K-Nearest Neighbors, Random Forest, and Multi-Layer Perceptron, were evaluated to identify algorithms that

balance accuracy, recall, and efficiency. Emphasizing URL-based features allows detection to occur before a user accesses a webpage, providing proactive protection against phishing threats.

3. Background

Machine learning has become a leading approach to detecting phishing websites by analyzing lexical and structural URL patterns. Early research by James, Sandhya, and Thomas (2013) demonstrated that models such as Support Vector Machines (SVM), Naïve Bayes, and K-Nearest Neighbors (KNN) can effectively classify phishing URLs using basic lexical and host-based features like domain length and number of subdomains. Basnet, Sung, and Liu (2014) expanded on this by examining the full "anatomy" of URLs, including lexical, host, and content-level features, and achieved error rates below 0.5%, proving that lightweight URL-based detection can perform in near real-time.

Further advances focused on improving accuracy and scalability. Sahingoz et al. (2019) combined lexical, statistical, and natural language processing (NLP) features, achieving nearly 98% accuracy and highlighting the benefits of integrating multiple feature types. Similarly, Gupta et al. (2021) proposed a real-time lexical-based approach using only nine lightweight URL features, reaching 99.57% accuracy and 99.7% precision with a Random Forest model while maintaining rapid classification speeds.

Most recently, Prasad and Chandra (2024) introduced the PhiUSIIL framework, which integrates a similarity index and incremental learning to adapt as new phishing patterns emerge. Their model achieved up to 99.8% accuracy and produced the PhiUSIIL Phishing URL Dataset, which contains both URL and webpage-level features for large-scale evaluation.

Collectively, these studies trace the evolution of phishing detection from early lexical models to adaptive, ensemble-based frameworks. The current research continues this progression by comparing multiple supervised learning algorithms on the PhiUSIIL dataset to identify models that balance high accuracy, recall, and efficiency for real-time phishing detection.

4. Dataset

The PhiUSIIL Phishing URL Dataset is a comprehensive dataset designed for phishing detection research. It contains 235,795 samples, divided into 134,850 legitimate and 100,945 phishing URLs. Each instance is characterized by 54 features, extracted from both URL structure (lexical features) and webpage source code (HTML features). The dataset is purely tabular, containing both numerical and categorical data.

- Dataset Composition Data Type: Numerical and Categorical
- Samples: 235,795 total (134,850 legitimate, 100,945 phishing)
- Features: 54 + 1 label column
- Label: 1 = legitimate, 0 = phishing
- Source: UCI Machine Learning Repository – PhiUSIIL Dataset

The data features could be split into 2 main categories (URL and Webpage) with some subcategories:

- URL and Domain Features

- **FILENAME** – Name of the file containing the URL. Not used for analysis.
- **URL** – The complete URL string.
- **URLLength** – Total number of characters in the URL. Longer URLs are sometimes suspicious.
- **Domain** – The domain name extracted from the URL.
- **DomainLength** – Number of characters in the domain. Extremely long domains may indicate phishing.
- **IsDomainIP** – Binary (1/0) indicating if the domain is an IP address instead of a domain name. IP addresses in URLs are often suspicious.
- **TLD** – Top-Level Domain (e.g., **.com**, **.org**). Some TLDs have higher phishing probabilities.
- **URLSimilarityIndex** – Measures similarity of the URL to known legitimate URLs. High similarity could indicate phishing attempts using look-alike URLs.
- **CharContinuationRate** – Frequency of consecutive repeating characters in the URL. High rates can indicate obfuscation.
- **TLDLegitimateProb** – Probability that a given TLD is associated with legitimate websites.
- **URLCharProb** – Probability of characters in the URL appearing in legitimate URLs (character distribution check).
- **TLDLength** – Length of the TLD string.
- URL Structure & Obfuscation Features
 - **NoOfSubDomain** – Number of subdomains in the URL. Multiple subdomains can be used to mimic legitimate domains.
 - **HasObfuscation** – Binary indicator of whether the URL uses obfuscation techniques (e.g., encoding, hex).
 - **NoOfObfuscatedChar** – Count of obfuscated characters in the URL.
 - **ObfuscationRatio** – Ratio of obfuscated characters to total characters in the URL.
 - **NoOfLettersInURL** – Count of alphabetic characters.
 - **LetterRatioInURL** – Ratio of letters to total URL length.
 - **NoOfDigitsInURL** – Count of digits in the URL. Phishing URLs often include many digits.
 - **DigitRatioInURL** – Ratio of digits to total URL length.
 - **NoOfEqualsInURL** – Count of **=** characters. Used in query parameters, excessive use can be suspicious.
 - **NoOfQMarkInURL** – Count of **?** characters. Multiple query strings can indicate phishing.
 - **NoOfAmpersandInURL** – Count of **&** symbols in URL parameters.
 - **NoOfOtherSpecialCharsInURL** – Count of other special characters (like **@**, **#**, **!**).
 - **SpacialCharRatioInURL** – Ratio of special characters to total URL length.
 - **IsHTTPS** – Binary indicator if the URL uses HTTPS. Non-HTTPS URLs are more suspicious.
- Webpage Source / HTML Features
 - **LineOfCode** – Total lines of HTML code on the page.
 - **LargestLineLength** – Length of the longest line of HTML code.
 - **HasTitle** – Binary indicator if the webpage has a **<title>** tag.
 - **Title** – The actual text of the **<title>** tag.
 - **DomainTitleMatchScore** – Score indicating how much the domain matches the title.
 - **URLTitleMatchScore** – Score indicating how much the URL matches the title.

- **HasFavicon** – Binary indicator if the page has a favicon. Phishing pages sometimes mimic favicons.
- **Robots** – Presence of `robots.txt` file. Legitimate websites usually have it.
- **IsResponsive** – Indicates if the website is responsive (adapts to screen size).
- Redirection & Navigation Features
 - **NoOfURLRedirect** – Number of redirects on the URL. Excessive redirects can indicate phishing.
 - **NoOfSelfRedirect** – Number of self-redirects (redirects to same domain).
 - **HasDescription** – Binary indicator if the page has meta description. Legitimate sites often include this.
 - **NoOfPopup** – Number of popup elements on the page.
 - **NoOfiFrame** – Number of `<iframe>` tags. Often used in phishing for embedding external content.
 - **HasExternalFormSubmit** – Binary if form submission is to an external domain. Suspicious behavior.
- Form & Interaction Features
 - **HasSocialNet** – Presence of social network links.
 - **HasSubmitButton** – Presence of submit buttons on forms.
 - **HasHiddenFields** – Presence of hidden input fields. Can be used maliciously.
 - **HasPasswordField** – Presence of password input fields. Phishing sites may mimic login forms.
- Content & Target-Specific Features
 - **Bank** – Binary indicating if content targets banking services.
 - **Pay** – Binary indicating if content targets payment platforms.
 - **Crypto** – Binary indicating if content targets cryptocurrency services.
 - **HasCopyrightInfo** – Binary if copyright information is present. Legitimate websites usually have it.
 - **NoOfImage** – Number of images on the page.
 - **NoOfCSS** – Number of CSS stylesheets linked.
 - **NoOfJS** – Number of JavaScript files linked.
- Reference & Linking Features
 - **NoOfSelfRef** – Number of self-referencing links.
 - **NoOfEmptyRef** – Number of empty `href` attributes. Suspicious or incomplete HTML.
 - **NoOfExternalRef** – Number of external links. Excessive external links can indicate phishing.

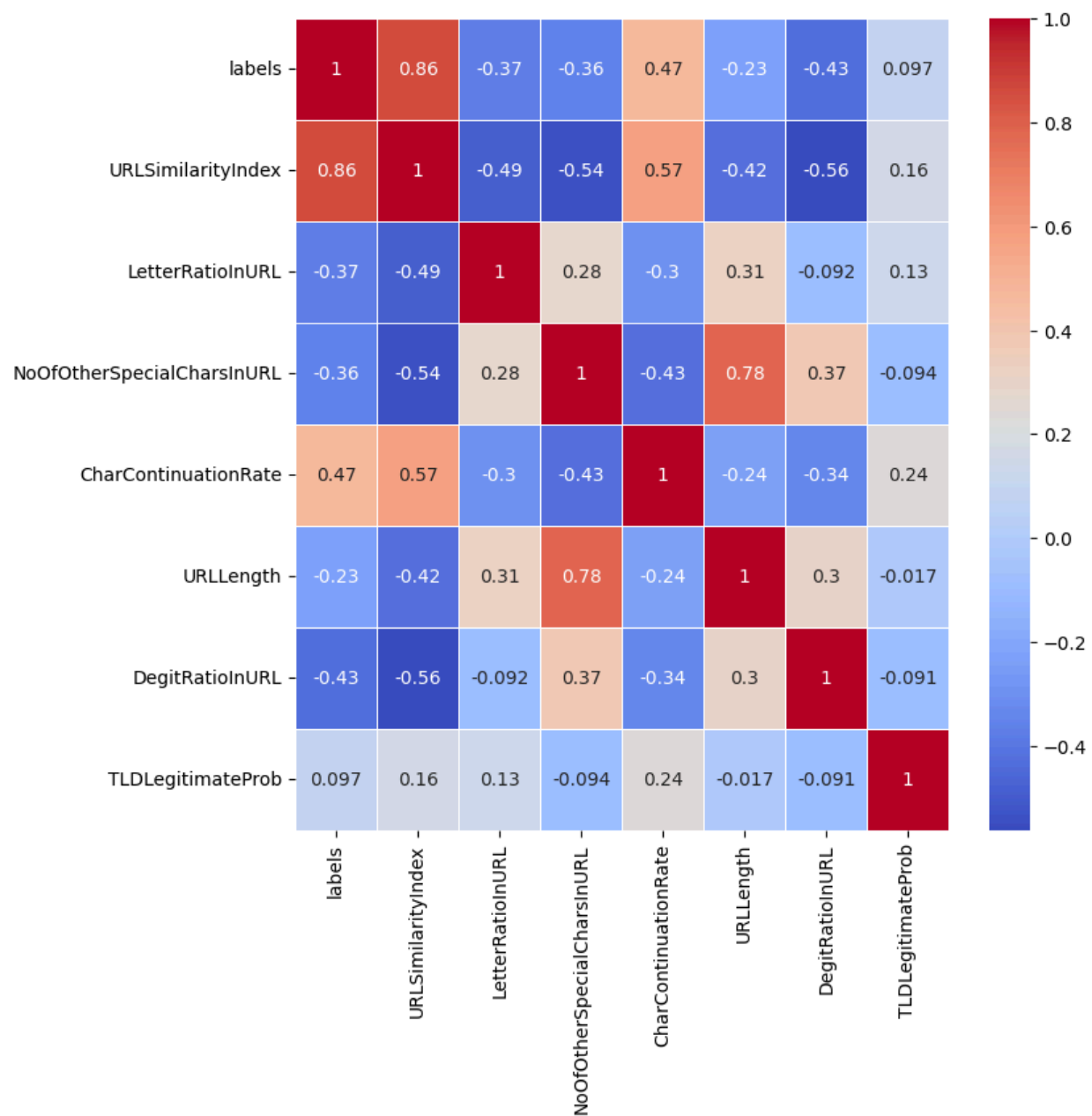


Figure 1: URL Features Correlation Matrix

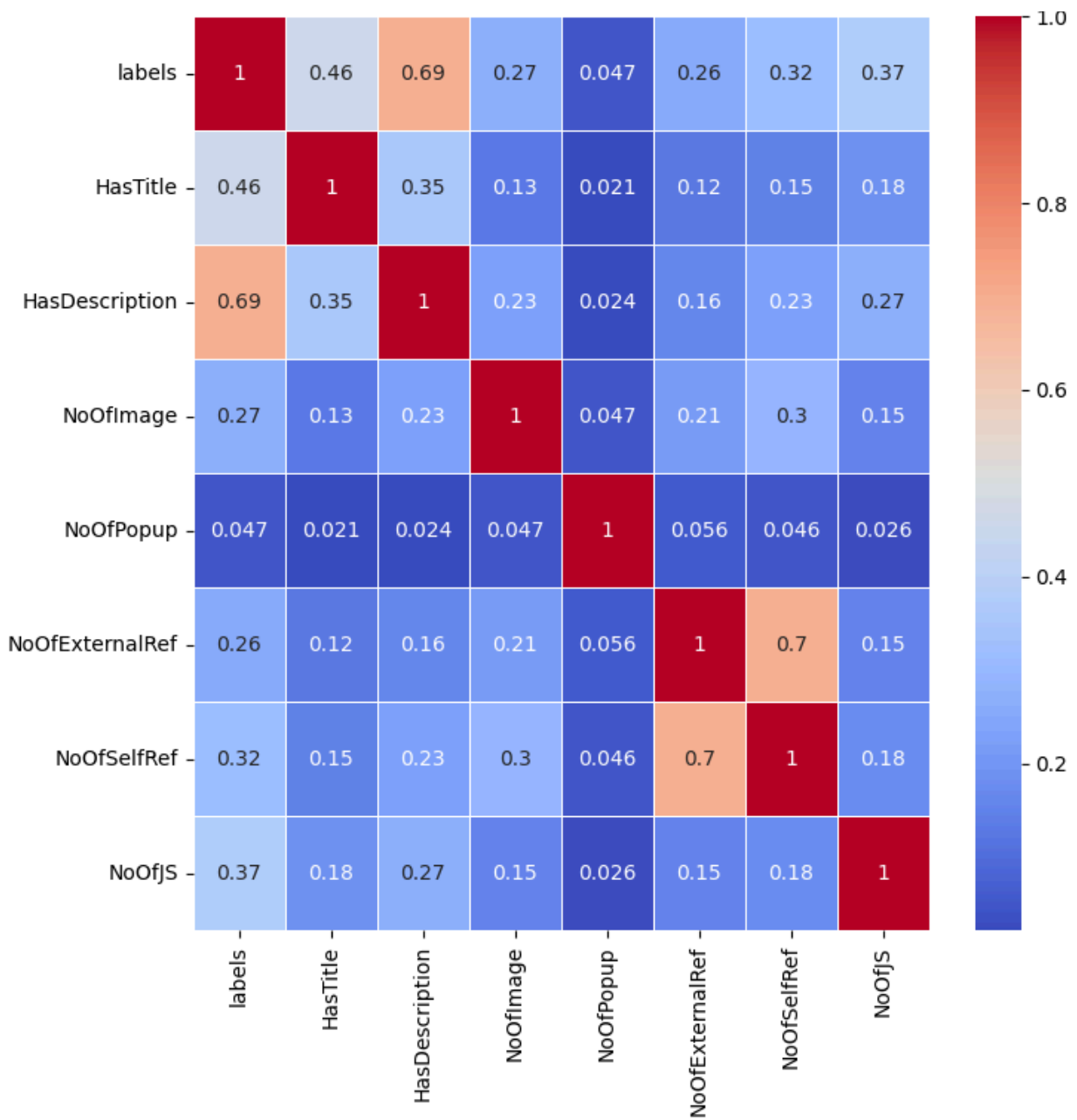


Figure 2: Webpage Features Correlation Matrix

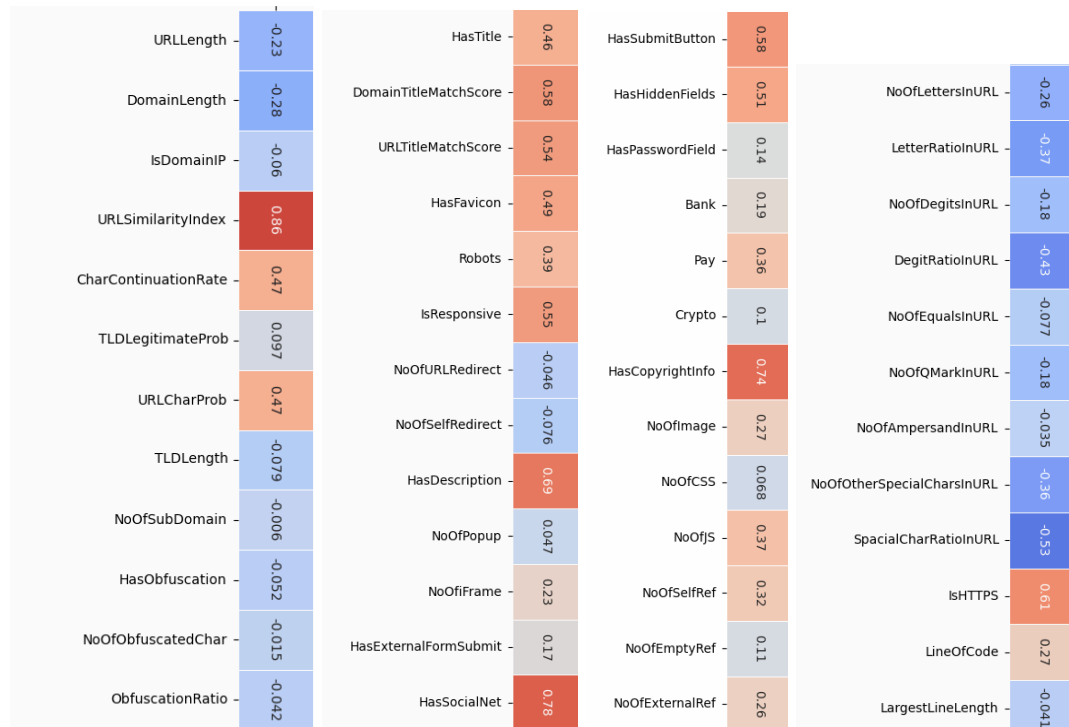


Figure 3: All Features Correlation w/ Labels

5. Methodology/Models

The goal of this research was to classify URLs as either **phishing** or **legitimate** based on their structural and content-based features. Because this is a **supervised binary classification problem**, machine learning algorithms were trained on labeled data where each sample was either a phishing (1) or legitimate (0) URL.

Data Preparation

The data was randomly split into **80% for training** and **20% for testing**, ensuring class balance. Prior to model training, non-informative attributes such as the file name and raw URL text were dropped. Features were standardized or normalized depending on the model (e.g., standard scaling for KNN and MLP).

Two main feature subsets were explored:

- **URL and Domain Features**
- **Webpage Source / HTML Features**

All models were implemented using **scikit-learn**, trained on both subsets separately to assess whether surface-level URL structure or deep website content better predicts phishing behavior.

Models Used

- **Logistic Regression:** Logistic Regression is a **linear model** that estimates the probability that an input belongs to a particular class. It applies the logistic (sigmoid) function to a linear combination of the input features. In phishing detection, this model captures linear relationships between URL characteristics (like digit ratio or TLD legitimacy probability) and the likelihood of a URL being phishing. It's simple, interpretable, and efficient, often serving as a strong baseline.
- **Decision Tree:** The Decision Tree model recursively splits the dataset based on features that provide the highest **information gain** (using Gini impurity or entropy). A maximum depth of 6 was chosen to reduce overfitting. This model is capable of learning **nonlinear feature relationships**; for instance, it may identify that a combination of "long URL + multiple subdomains + missing HTTPS" strongly predicts phishing. Its transparency allows visualization of key decision rules used for classification.
- **K-Nearest Neighbors:** The KNN algorithm classifies a URL based on the **majority label among its k nearest neighbors** in the feature space. A value of 10 was chosen for k. Distances are typically computed using the **Euclidean distance metric**. KNN does not explicitly train parameters; instead, it relies on **feature similarity**, which works well when features are properly scaled. For phishing URLs, this means that a URL structurally similar to known phishing examples (e.g., with similar length, character ratios, or special symbol frequency) will likely also be labeled as phishing.
- **Random Forest:** The Random Forest is an **ensemble learning method** that constructs multiple Decision Trees and aggregates their outputs via **majority voting**. Each tree is trained on a random subset of features and data samples (bagging). This diversity among trees allows the model to achieve **high accuracy and robustness** while minimizing overfitting. In phishing detection, Random Forests are highly effective because they capture complex, nonlinear dependencies, such as the interplay between URL obfuscation ratios and HTML-level cues like missing metadata.
- **Multi-Layer Perceptron (MLP Classifier):** The MLP is a **feedforward neural network** consisting of an input layer, one or more hidden layers, and an output layer. The hidden layers use nonlinear activation functions (typically ReLU or tanh), enabling the network to model complex relationships between features. The chosen architecture (10, 5) represents two hidden layers with 10 and 5 neurons, respectively. The MLP was trained using backpropagation with an Adam optimizer. It is especially powerful for detecting **subtle, nonlinear patterns** in phishing URLs that other models might overlook.

Evaluation Metrics

Models were evaluated using:

- **Accuracy:** proportion of total correct predictions.
- **Precision:** ratio of correctly identified phishing URLs among all predicted phishing URLs.
- **Recall (Sensitivity):** ratio of correctly identified phishing URLs among all actual phishing URLs.
→ High recall indicates the model rarely misses a phishing site.
- **F1 Score:** harmonic mean of precision and recall, balancing both false positives and false negatives.

Given the real-world importance of identifying phishing URLs, **recall** was prioritized; a false negative (failing to flag a phishing site) is far more dangerous than a false positive (incorrectly labeling a legitimate site as phishing).

6. Results and Discussion

Confusion matrix

A confusion matrix shows the performance of a model by comparing the predicted labels with the actual labels. It shows the correct and incorrect predictions in 4 categories: true phishing (correctly predicted phishing), false phishing (legitimate URLs labeled as phishing incorrectly), false legitimate (phishing URLs labelled as legitimate incorrectly) and true legitimate (correctly predicted legitimate URLs). They are organized as shown below

[True Phishing, False Legitimate]

[False Phishing, True Legitimate]

Classifier	Confusion Matrix	Accuracy	Precision	Recall	F1
Logistic Regression	[19937, 187] [2, 27033]	99.599%	99.313%	99.993%	99.652%
Decision Tree (max depth 6)	[19996, 128] [0, 27035]	99.729%	99.529%	100%	99.764%
K Nearest Neighbors (k=10)	[20073, 51] [0, 27035]	99.892%	99.812%	100%	99.906%
Random Forest(default)	[20087, 37] [0, 27035]	99.922%	99.863%	100%	99.932%
MLP Classifier (10,5)	[20034, 90] [0, 27035]	99.809%	99.668%	100%	99.834%

Table 1: Performance of models with **URL features as inputs**: ['URLSimilarityIndex', 'LetterRatioInURL', 'NoOfOtherSpecialCharsInURL', 'CharContinuationRate', 'URLLength', 'DegitRatioInURL', 'TLDLegitimateProb']

Interpretation:

All models achieved outstanding performance on the URL-only feature subset. The Random Forest achieved the **highest accuracy (99.922%)**, while maintaining perfect recall.

The **100% recall** observed in several models means that **every phishing URL in the test set was correctly identified as phishing**, i.e., **zero false negatives**. This is critical for cybersecurity; the system never failed to detect a phishing attempt.

The slight variations in precision (99.3–99.8%) indicate a few false positives (legitimate URLs incorrectly flagged). However, these errors are tolerable in security contexts, where it is preferable to overflag than to miss threats.

Among interpretable models, Decision Tree provided insight into feature importance: *URLLength*, *DegitRatioInURL*, and *TLDLegitimateProb* were consistently ranked as top predictors. These results suggest that **URL structure alone** provides a remarkably strong signal for detecting phishing activity.

For Website features:

Model	Parameters	Accuracy	Precision	Recall	F1
Logistic Regression	[19888, 236] [600, 26435]	98.227%	99.115%	97.781%	98.443%
Decision Tree	[19878, 246] [250, 26785]	98.948%	99.090%	99.075%	99.083%
Nearest Neighbors	[19943, 181] [285, 26750]	99.019%	99.328%	98.946%	99.136%
Random Forest	[19945, 179] [134, 26901]	99.336%	99.339%	99.504%	99.422%
MLP Classifier	[19837, 287] [181, 26854]	99.008%	98.943%	99.330%	99.136%

Table 2: Performance of models with **Website features as inputs**: ['HasTitle', 'HasDescription', 'NoOfImage', 'NoOfPopup', 'NoOfExternalRef', 'NoOfSelfRef', 'NoOfJS']

Interpretation:

Performance using website-level HTML features was slightly lower than with URL-only features. This is likely due to higher noise and variability in webpage structures. Legitimate websites can vary greatly in code complexity, pop-ups, and embedded elements. Nevertheless, the **Random Forest** again outperformed other models, achieving **99.336% accuracy** and **99.504% recall**, confirming its reliability across different feature types.

The MLP and KNN models also performed competitively, suggesting that nonlinear feature interactions (such as between “NoOfPopup” and “HasDescription”) play a role in phishing identification. Interestingly, features like **HasTitle**, **HasDescription**, and **NoOfExternalRef** emerged as top contributors, reinforcing that missing metadata and excessive external linking are common in phishing sites.

General Discussion

Across both feature subsets, **ensemble and nonlinear models** (Random Forest, KNN, and MLP) consistently achieved superior performance, demonstrating that phishing URLs and webpages exhibit **complex, nonlinear patterns** rather than purely linear relationships.

The results indicate that **structural URL analysis alone is sufficient** for highly accurate phishing detection, which is a valuable finding for real-time systems that can flag suspicious links before users even load the webpage, ensuring immediate safety.

However, perfect recall does not necessarily imply flawless generalization. Future work should test these models on **completely unseen URLs** from different time periods or domains to ensure robustness against evolving phishing tactics.

7. Conclusion

This study demonstrates that machine learning can accurately detect phishing URLs using both URL structure and webpage features. URL-based features alone provide extremely high predictive power, enabling phishing detection before a user visits a website. The Random Forest model achieved 99.922 percent accuracy and 100 percent recall with URL-only features, ensuring that all phishing URLs were correctly identified. Webpage features also enhanced detection, though with slightly lower accuracy.

These results confirm that analyzing the structural properties of URLs is sufficient for highly effective phishing detection, making real-time, preemptive protection feasible. Future work could include testing the models on new and evolving phishing datasets, integrating adaptive feature updates, and optimizing neural network architectures to improve speed and scalability. Overall, this research highlights the potential for efficient phishing prevention systems that proactively protect users from malicious websites.

8. Acknowledgements

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9. References

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